

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

50% task-completion time horizon doubles every ~7 months ($R^2 = 0.97$)

A New Metric: 50% Task-Completion Time Horizon



The 50% Task-Completion Time Horizon:

The duration of tasks (measured by how long they take skilled human professionals) that an AI model can complete with 50% success rate. This provides an intuitive, human-grounded measure of AI capability across vastly different models.

[Methodology overview: figure_2.jpg]

figure_2.jpg

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST
97 tasks
Diverse software tasks
1 min – 30 hrs

RE-Bench
7 tasks
Difficult ML research
engineering tasks (8 hrs each)

SWAA
66 tasks
Short single
1 sec – 30 min

TOTAL
170 Tasks
2,529 human hrs

Example Tasks by Duration:

Task	Duration	Description
find_shell_script	3 seconds	Multiple choice: Which file is a shell script?
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input
munge_data	56 minutes	Write Python script to transform JSON data by inferring rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of backtesting tool

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~2 Hours (o3) in 6 Years

~207 days

Doubling Time

$R^2 = 0.97$

Excellent Fit

7 months

Per Doubling

figure_1.jpg

Exponential Trend Chart

Key Milestones

GPT-2 (2019)	~2 sec
GPT-3 (2020)	~9 sec
GPT-4 (2023)	~5 min
Claude 3.5 Sonnet	~28 min
o1 (2024)	~39 min
o3 (2025)	~1h 50m

Success Probability Drops Sharply as Task Length Increases

Per-model success curves show frontier models push the curve rightward — completing longer tasks successfully

figure_3.jpg

Per-Model Success Probability Curves

Key Insight

GPT-2 (2 sec): Limited to shortest tasks only

GPT-4 (5 min): Expands to multi-minute tasks

Claude 3.7 Sonnet (54 min): Significant rightward shift

o3 (1h 50m): Approaching 2-hour frontier

All models show steep success drop-off as task length increases

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Improved Logical Reasoning

Better chain-of-thought reasoning enables tackling complex multi-step problems

Better Tool Use

Enhanced ability to interact with files, APIs, and external systems

Greater Reliability

More consistent task completion with fewer failures and dead-ends

Error Adaptation

Improved self-awareness and ability to recover from mistakes

Limitation:

Performance is notably lower on less structured, "messier" tasks. External validity remains an open question — tasks may not perfectly represent the average segment of real-world intellectual labor.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Naive Extrapolation

If the trend continues, AI reaches >1 month time horizon (167 work hours) between mid-2028 and mid-2031

Today (o3)

~2 hrs

167 hrs

Acceleration Signal

2023–2025 growth rate is ~20% faster than the 2019–2025 rate, suggesting possible acceleration since 2024

The 80% time horizon shows similar trend but with horizons ~5x shorter

External Validity

Tasks may not perfectly represent real-world intellectual labor distribution

Trend Changes

Performance could slow down or speed up unexpectedly

Task Selection

Benchmark tasks are curated — may not reflect average real-world work

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1

Robust Metric

50% time horizon provides a robust, intuitive metric for tracking AI autonomy with $R^2=0.97$ over 6 years

2

Doubling Trend

Time horizon doubles every ~7 months with possible acceleration since 2024

3

Current Frontier

o3 achieves ~2 hours of human-equivalent work — up from 2 seconds for GPT-2 in 2019

4

External Validity

Open question whether benchmark tasks perfectly represent real-world intellectual labor

5

Key Limitations

Poor performance on unstructured tasks; uncertain generalization to real-world job distributions